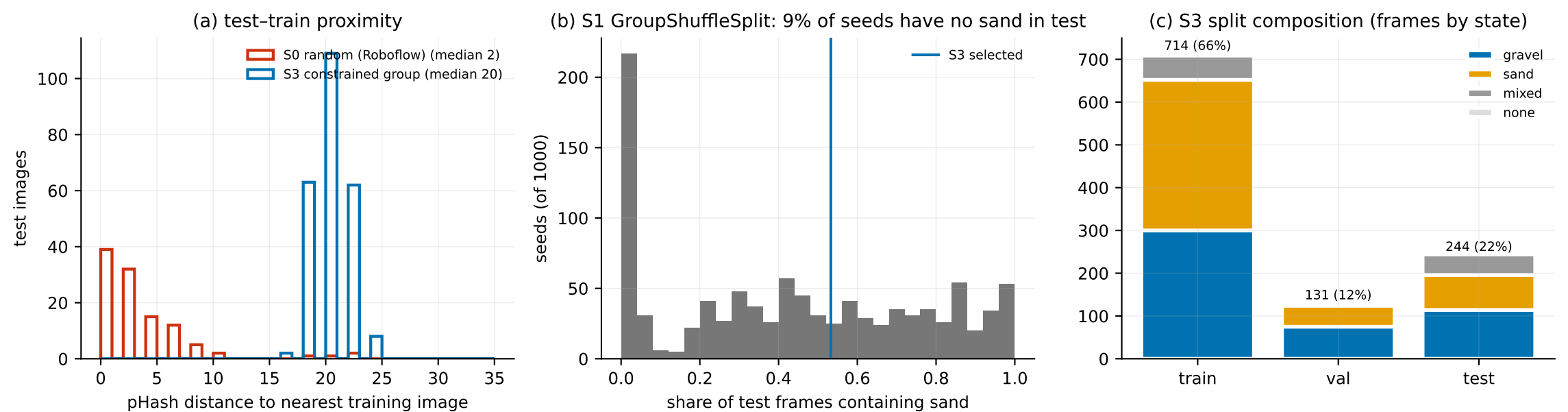
(a) test–train proximity
S0 random (Roboflow) (median 2)
S3 constrained group (median 20)
test images
pHash distance to nearest training image
(b) S1 GroupShuffleSplit: 9% of seeds have no sand in test
S3 selected
seeds (of 1000)
share of test frames containing sand
(c) S3 split composition (frames by state)
gravel
sand
mixed
none
714 (66%)
131 (12%)
244 (22%)
train
val
test